



MEDICAL COST ANALYSIS

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AGENDA

- Problem Statement
- Dataset Overview and Tools
- Dashboards
- Hypothesis Testing
- Insights
- Contact Info



PROBLEM STATEMENT



Problem Statement: What are the key factors that significantly impact medical costs for individuals?

Objective: Perform an exploratory data analysis to identify which factors (e.g., age, BMI, smoking status) are most strongly correlated with high medical costs.

DATASET OVERVIEW AND TOOLS



About Dataset : Dataset Contains Following Attributes

age: age of primary beneficiary

sex: insurance contractor gender, female, male

bmi: Body mass index, providing an understanding of body, weights that are relatively high or low relative to height, objective index of body weight (kg / m^2) using the ratio of height to weight, ideally 18.5 to 24.9

children: Number of children covered by health insurance / Number of dependents

smoker: Smoking

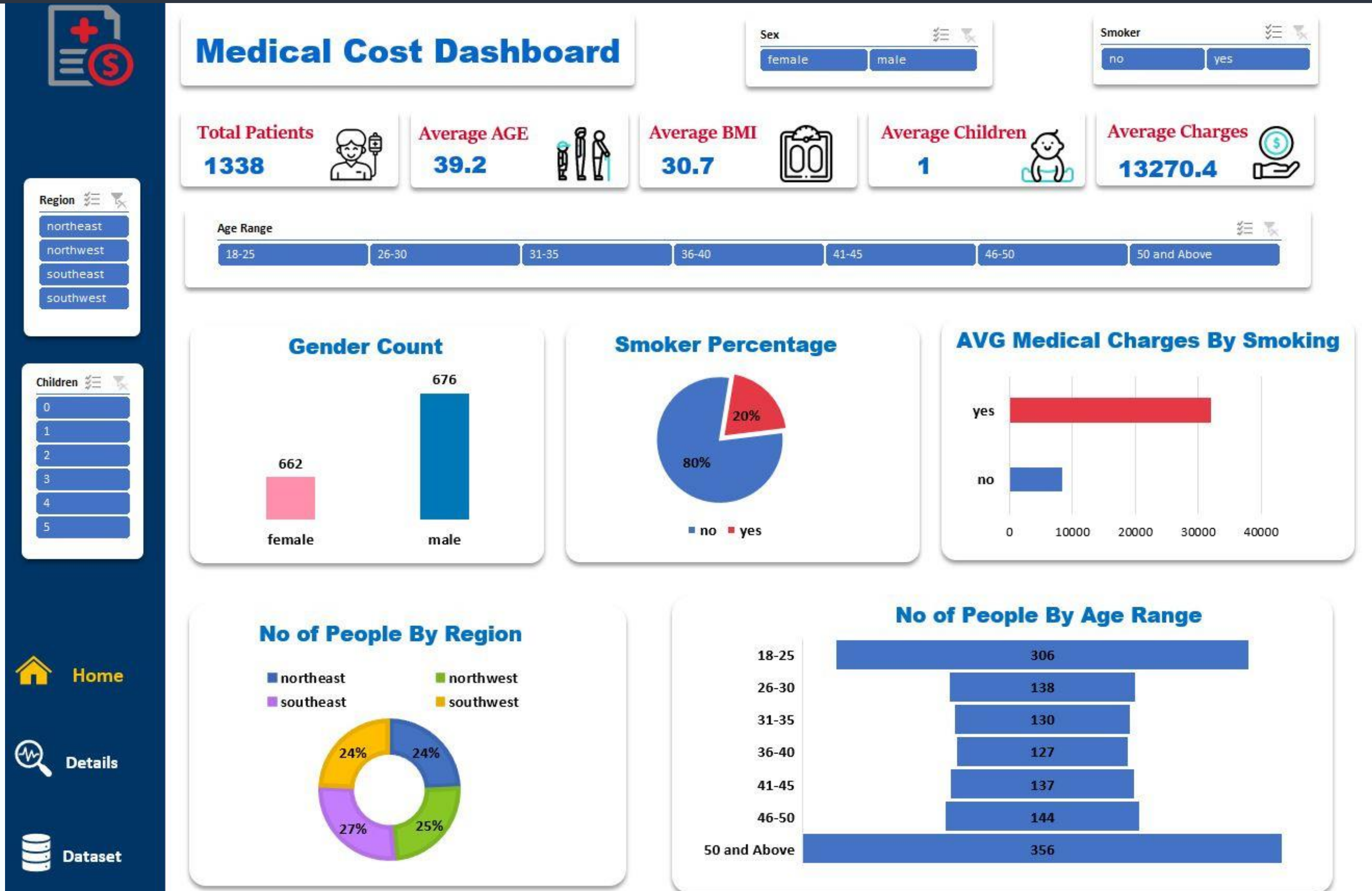
region: the beneficiary's residential area in the US, northeast, southeast, southwest, northwest.

charges: Individual medical costs billed by health insurance



Tools: These tools have used to Perform the Data Analysis and Project creation Excel , SQL , Python .

DASHBOARDS



DASHBOARDS



Medical Cost Dashboard

Sex

female

male

Smoker

no

yes

Total Patients

1338



Average AGE

39.2



Average BMI

30.7



Average Children

1



Average Charges

13270.4



Region

northeast

northwest

southeast

southwest

Children

0

1

2

3

4

5

Age Range

18-25

26-30

31-35

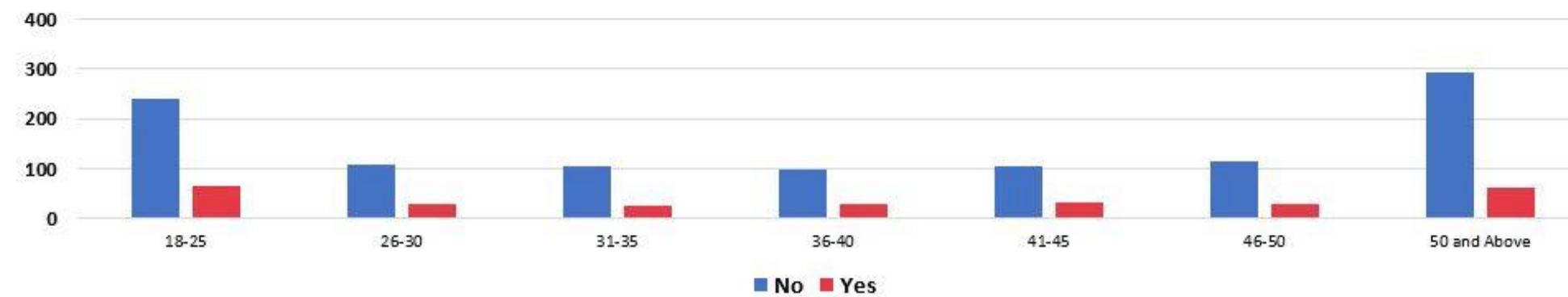
36-40

41-45

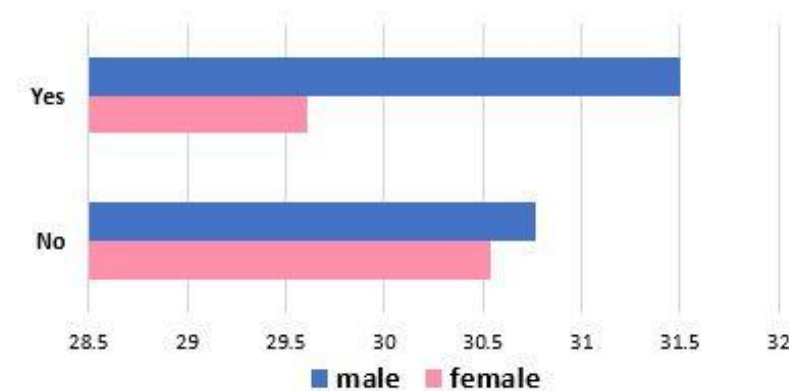
46-50

50 and Above

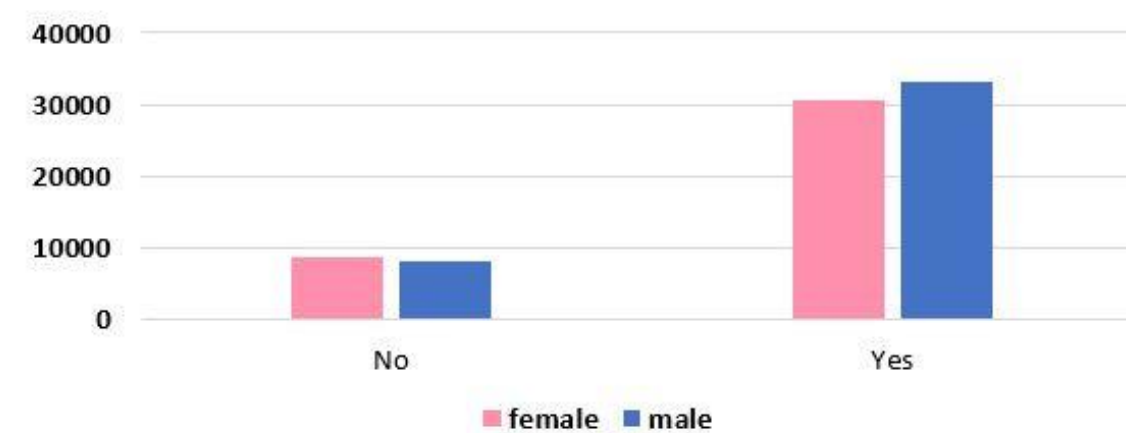
No of Smokers By Age Range



AVG BMI By Gender and Smoking Status



AVG Medical Charges By Gender and Smoking Status



Home



Details



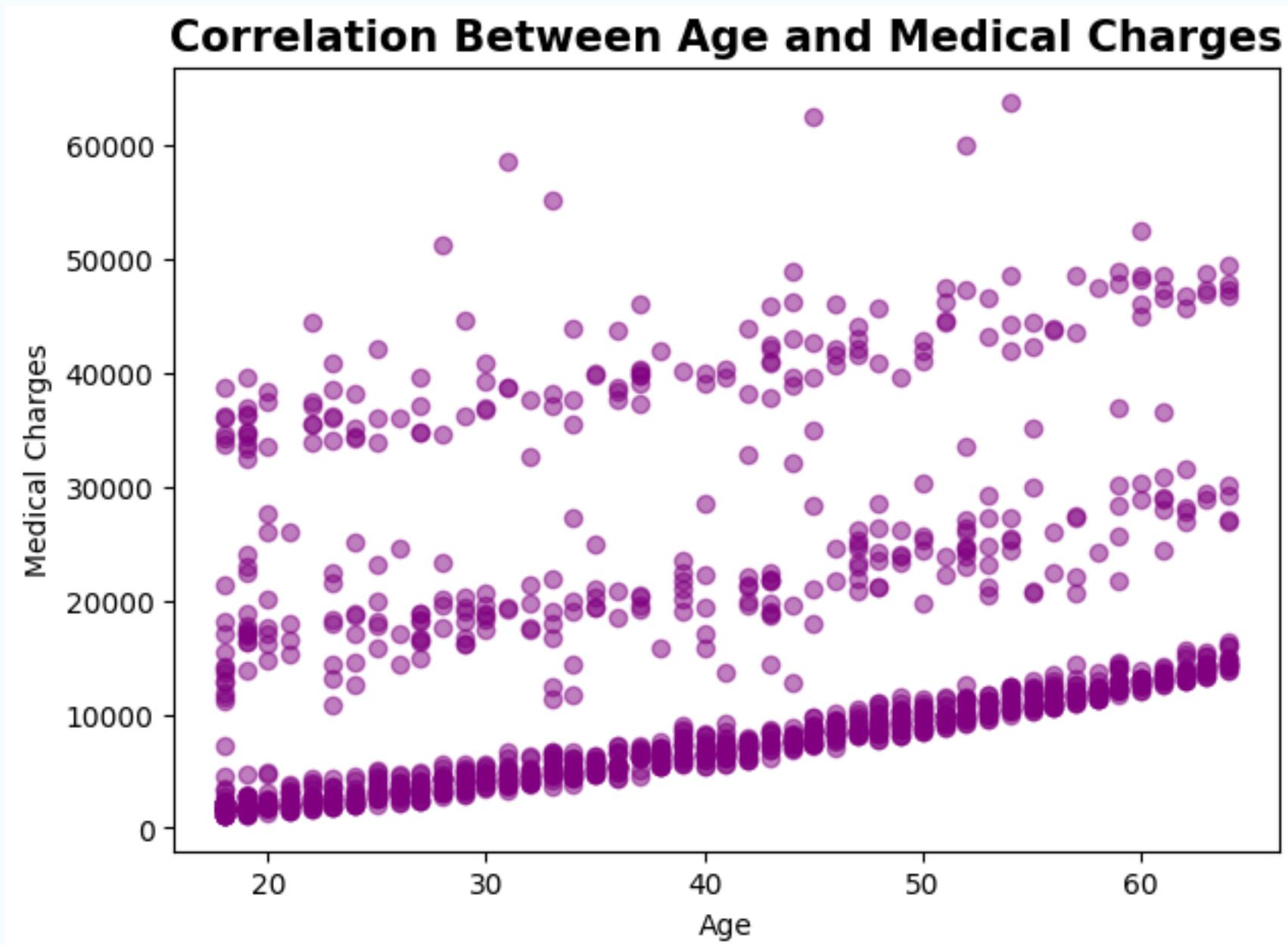
Dataset

HYPOTHESIS TESTING

Performed Correlation Pearson Test

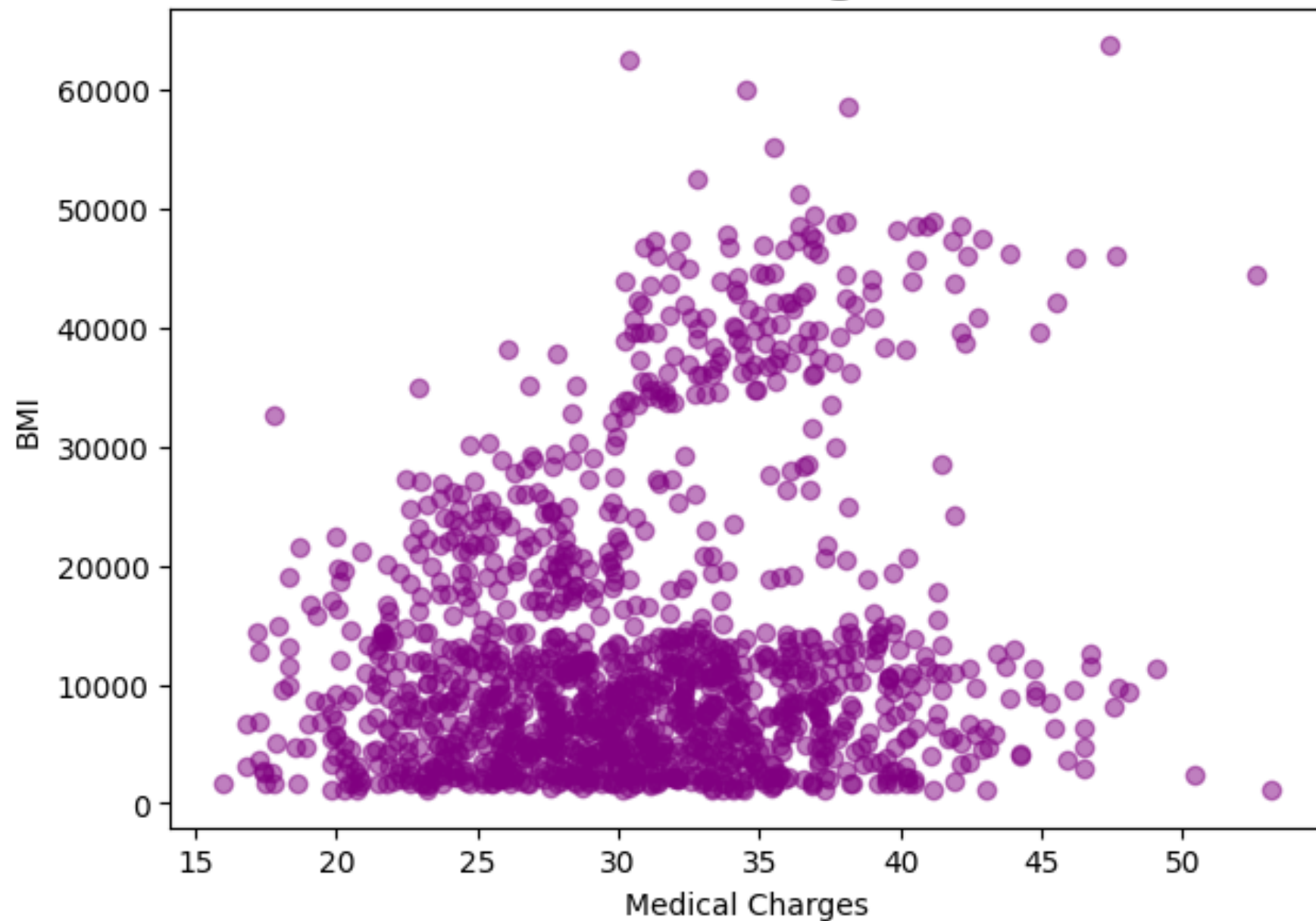
Pearson correlation: 0.2990081933306476, p-value: 4.886693331718663e-29
There is a significant correlation between Age and Medical Charges.

**There is a Little Positive Correlation
between Age and Medical Charges . If the
Age will increase there is a chance the
medical charges of a person will also
increase**



HYPOTHESIS TESTING

BMI and Medical Charges Correlation



Performed Correlation Pearson Test

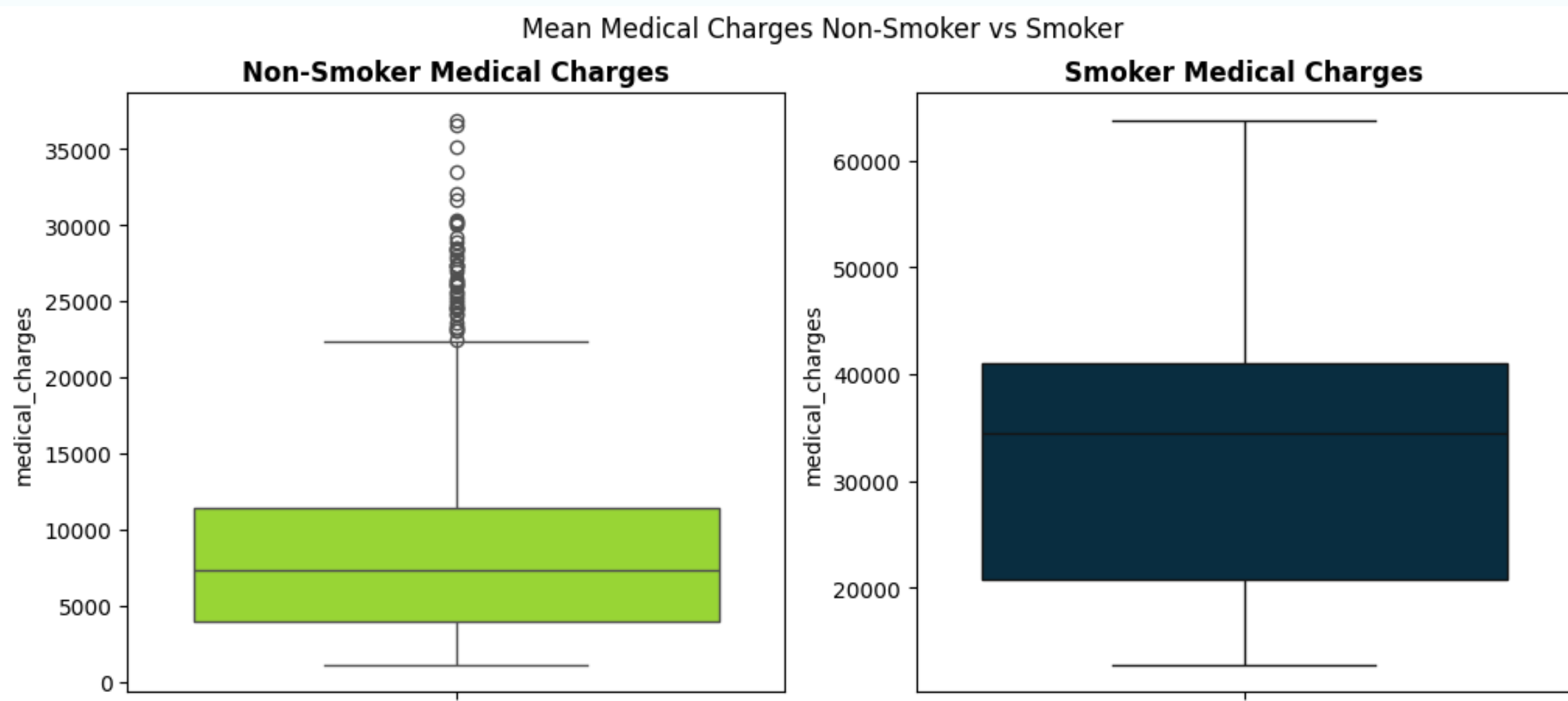
Pearson correlation: 0.1983409688336289, p-value: 2.459085535116776e-13
There is a significant correlation between BMI and Medical Charges.

There A positive correlation between BMI and Medical Charges . We can observe the linear relationship between both. Whenever BMI will increase Medical Charges of Person will also increase

HYPOTHESIS TESTING

**Performed T-Test to compare
Mean of Smoker vs Non-Smoker
Medical Charges**

**By Seeing the figures and by observing
the test we can say that there is a
significant difference between mean
of medical charges of smoker and
non-smoker**

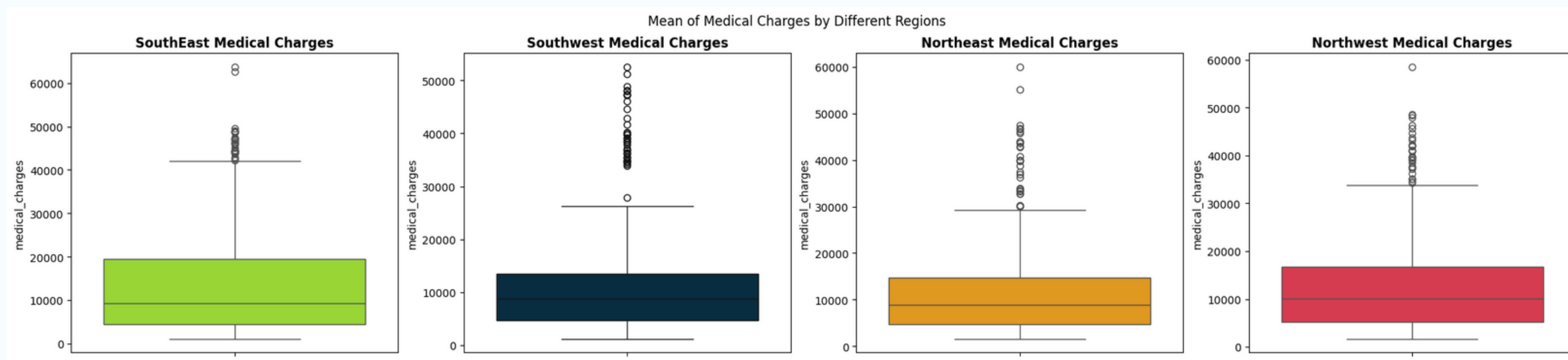


`5.88946444671698e-103`

`There is a Difference Between Mean of Medical Charges of Smoker and Non-Smoker`

HYPOTHESIS TESTING

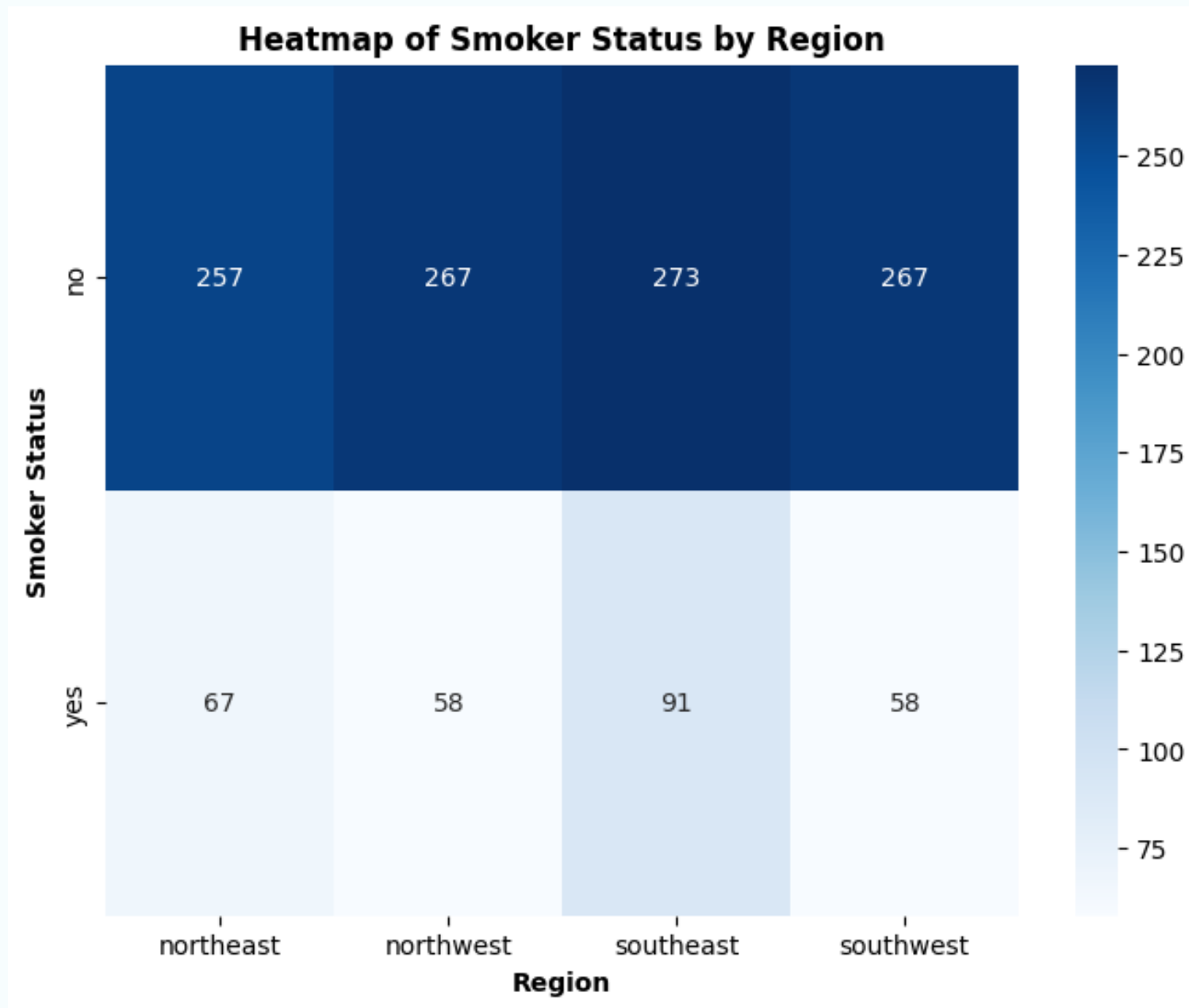
Performed One-way ANNOVA to compare
Mean Medical Charges Across different
regions



By Seeing the figures and by observing
the test we can say that there is a
significant difference between mean
of medical charges of different
Regions

```
ANOVA F-statistic: 2.96962669358912, p-value: 0.0308933560705201
There is a significant difference in medical charges across regions.
```

HYPOTHESIS TESTING



Performed Chi-Square Test to check association between smoker and region

By observing the test we can say that there is no significant Association between Smoker Status and Region

Chi-square test statistic: 7.34347776140707, p-value: 0.06171954839170547
There is no significant association between smoker status and region.

INSIGHTS

- Males slightly outnumber females in the dataset, with **676** males and **662** females.
 - There are **20%** Smokers and **80%** Non-Smokers of the Whole population of the patients
- There is a Significantly High Difference Between The Medical Charges of Patients who smoke and who don't smoke
- Northeast Patients are **24%** Northwest Patients are **24%** . Southeast Patients are **27%** and Southwest Patients are **25%** of the Whole Population
 - Most of The People are of Age Range **18-25** and **50 above**

INSIGHTS

- Most of the Smokers and Non-Smokers are of Age Range **18-25** and **50 above** other are less numbers of smokers and non-smokers .
- Male Smokers have higher BMI as Compared to Female Smokers and also Male Non-smokers have higher BMI as compared to Female Non-Smokers
- Average Medical Charges of Both Males and Females are high If they are smokers as compared to Non-Smokers
 - Further Insights can be Derived by Using the Dashboard

THANK YOU



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